Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Air is pumped into an automobile tube upto a pressure of $200\mathrm{kPa}$ in the morning when the air temperature is $22^{\circ}\mathrm{C}$. During the day, temperature rises to $42^{\circ}\mathrm{C}$ and the tube expands by 2%. The pressure of the air in the tube at this temperature, will be approximately

(A) $212\mathrm{kPa}$

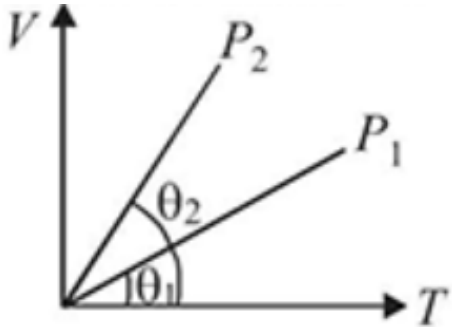
(B) $209\mathrm{kPa}$

(C) $206\mathrm{kPa}$

(D) $200\mathrm{kPa}$

Q.2

The figure shows the volume V versus temperature T graphs for a certain mass of a perfect gas at two constant pressures of P_1 and P_2. What inference can you draw from the graphs?



(A) $\mathrm{P}_1 > \mathrm{P}_2$

(B) $\mathrm{P}_1 < \mathrm{P}_2$

(C) $P_1 = P_2$

(D) No inference can be drawn due to insufficient information.

Q.3

A graph is plotted with $\mathrm{PV/T}$ on y -axis and mass of the gas along x-axis for different gases. The graph is

(A) a straight line parallel to x -axis for all the gases

(B) a straight line passing through origin with a slope having a constant value for all the gases

(C) a straight line passing through origin with a slope having different values for different gases

(D) a straight line parallel to y-axis for all the gases

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Q.4

A sample of an ideal gas occupies a volume V at a pressure P and absolute temperature T. The mass of each molecule is m. The equation for density is

(A) mkT

(B) $\mathrm{P/kT}$

(C) $\mathrm{P/(kTV)}$

(D) $\mathrm{Pm/kT}$

Q.5

A vessel has $6\ \mathrm{g}$ of hydrogen at pressure P and temperature $500\ \mathrm{K}$. A small hole is made in it so that hydrogen leaks out. How much hydrogen leaks out if the final pressure is $\mathrm{P}/2$ and temperature falls to $300\ \mathrm{K}$?

(A) $2\ \mathrm{g}$

(B) $3\ \mathrm{g}$

(C) $4\ \mathrm{g}$

(D) $1\ \mathrm{g}$

Q.6

When do real gases approach the ideal gas behaviour?

(A) At low pressure and low temperature

(B) At low pressure and high temperature

(C) At high pressure and high temperature

(D) At high pressure and low temperature

Q.7

Consider a collection of a large number of dust particles each with speed v. The direction of velocity is randomly distributed in the collection. What is the magnitude of the relative velocity between a pairs in the collection?

(A) $\frac{3v}{\pi}$

(B) $\frac{4v}{\pi}$

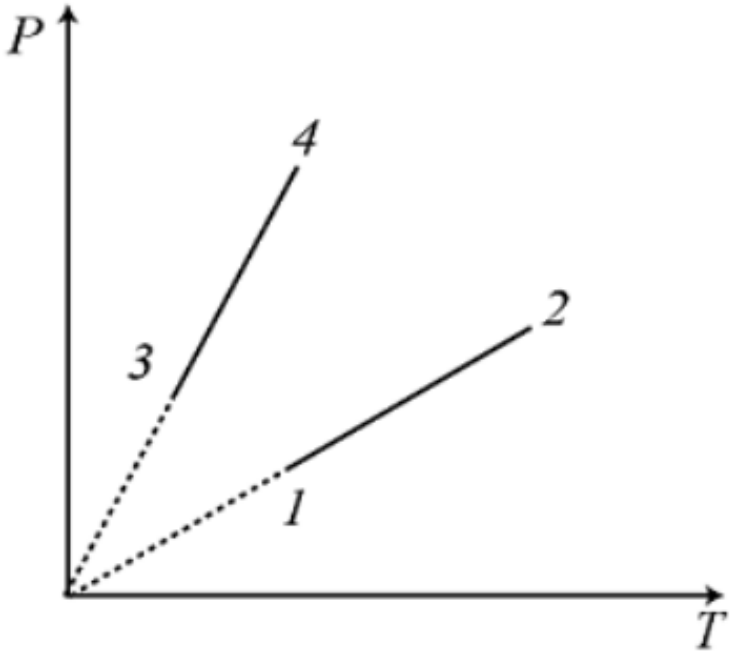
(C) $\frac{2v}{\pi}$

(D) $\frac{v}{\pi}$

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Q.8

Pressure versus temeprature graph of an ideal gas of equal number of moles of different volumes are plotted as shown in figure. Choose the correct alternative



(A) $V_1 = V_2,\ V_3 = V_4$ and $V_2 > V_3$

(B) $V_1 = V_2,\ V_3 = V_4$ and $V_2 < V_3$

(C) $V_1 = V_2 = V_3 = V_4$

(D) $V_4 > V_3 > V_2 > V_1$

Q.9

Two containers A and B are partly filled with water and closed. The volume of A is twice that of B and it contains half the amount of water in B. If both are at the same temperature, the water vapour in the containers will have pressure in the ratio of

(A) $1 : 2$

(B) $1 : 1$

(C) $2 : 1$

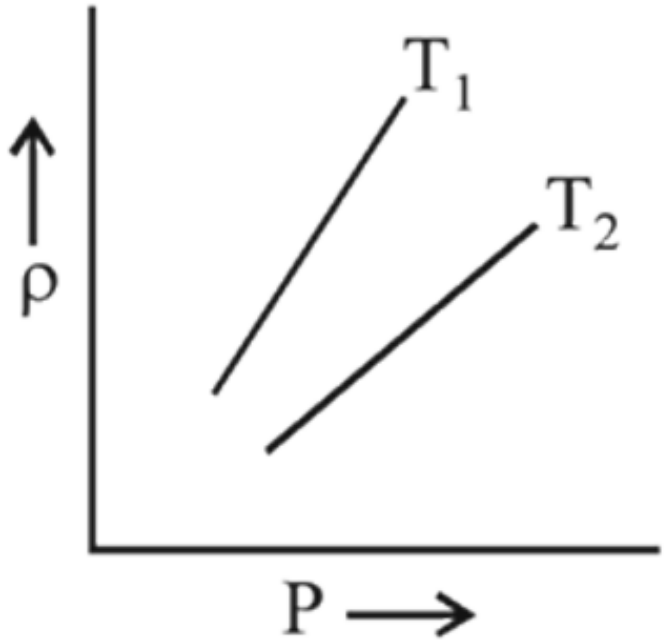
(D) $4 : 1$

Q.10

The density (ρ) versus pressuure (P) of a given mass of an ideal gas is shown at two temperatures T_1 and T_2

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Then relation between T_1 and T_2 may be



(A) $T_1 > T_2$

(B) $T_2 > T_1$

(C) $T_1 = T_2$

(D) All the three are possible

Q.11

Boyle' law is applicable for an

(A) adiabatic process.

(B) isothermal process.

(C) isobaric process.

(D) isochoric process

Q.12

A balloon contains $1500\ m^3$ of helium at $27^\circ C$ and 4 atmospheric pressure. The volume of helium at $-3^\circ C$ temperature and 2 atmospheric pressure will,

(A) $1500\ m^3$

(B) $1700\ m^3$

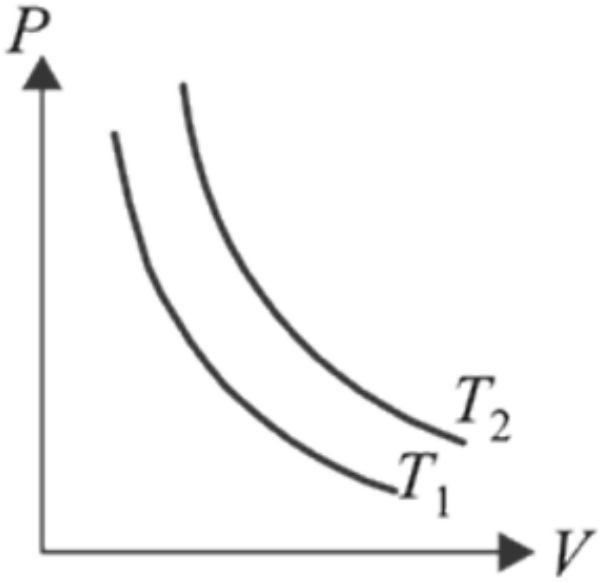
(C) $1900\ m^3$

(D) $2700\ m^3$

Q.13

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The figure shows graph of pressure and volume of a gas at two different temperatures T_1 and T_2. Which of the following inferences is correct?



(A) $T_1 > T_2$

(B) $T_1 = T_2$

(C) $T_1 < T_2$

(D) None of these

Q.14

At 10°C the value of the density of a fixed mass of an ideal gas divided by its pressure is x. At 110°C this ratio is:

(A) $\frac{383}{283}x$

(B) x

(C) $\frac{10}{110}x$

(D) $\frac{283}{383}x$

Q.15

We have $0.5\ \text{g}$ of hydrogen gas in a cubic chamber of side $3\ \text{cm}$ kept at NTP. The gas in the chamber is compressed keeping the temperature constant till a final pressure of $100\ \text{atm}$. What is the final volume of the gas after compression, assuming ideal gas law is valid? (Hydrogen molecules can be considered as spheres of radius $1\mathring{A}$).

(A) $2.9 \times 10^{-7}\ \text{m}^3$

(B) $6.7 \times 10^{-7}\ \text{m}^3$

(C) $2.7 \times 10^{-7}\ \text{m}^3$

(D) $4.7 \times 10^{-7}\ \text{m}^3$

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Q.16

A gas at NTP is suddenly compressed to one fourth of its original volume. If γ is supposed to be $3 / 2,$ then the final pressure is:-

(A) 4 atmosphere

(B) $\frac{3}{2}$ atmosphere

(C) 8 atmosphere

(D) $\frac{1}{4}$ atmosphere

Q.17

A truck moves with a constant speed. It is loaded with cooking gas containers. How will the temperature of the gas change?

(A) Increase

(B) Decrease

(C) Remains the same

(D) Decrease for some and increase for others

Q.18

How is the temperature of an ideal gas affected if it is kept in a closed container and transferred to a train moving with high speed?

(A) Increase.

(B) Decrease.

(C) Remains constant.

(D) May increase or decrease, depending on the nature of the gas.

Q.19

Internal energy of n_1 moles of monoatomic molecule of helium is same as the internal energy of n_2 moles of diatomic molecule of carbon monoxide at same temperature. Determine the ratio $n_1 : n_2$
Select the nearest option.

(A) 1.72

(B) 1.56

(C) 1.67

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(D) 1.69

Q.20

The molecules of two gases have radii in ratio $\frac{3}{2}$. Assuming that both the gases are under identical conditions and have equal volumes, calculate the ratio of the mean free paths of the molecules.

(A) 00.22

(B) 00.33

(C) 00.44

(D) 00.55

Answers & Solutions

Q.1 Answer:

$209\mathrm{kPa}$

Solution:

$$\frac{\mathrm{P_1\,V_1}}{\mathrm{T_1}} = \frac{\mathrm{P_2\,V_2}}{\mathrm{T_2}} \quad \text{Here, } \mathrm{P_1} = 200\mathrm{kPa}$$
$$\mathrm{T_1} = 22^{\circ}\mathrm{C} = 295\ \mathrm{K} \quad \mathrm{T_2} = 42^{\circ}\mathrm{C} = 315\ \mathrm{K}$$
$$\mathrm{V_2} = \mathrm{V_1} + \frac{2}{100}\,\mathrm{V_1} = 1.02\,\mathrm{V_1}$$
$$\therefore \quad \mathrm{P_2} = \frac{200 \times 315\,\mathrm{V_1}}{295 \times 1.02\,\mathrm{V_1}} = 209.37\mathrm{kPa}$$

Q.2 Answer:

$\mathrm{P_1} > \mathrm{P_2}$

Solution:

$\because \theta_1 < \theta_2 \Rightarrow \tan\theta_1 < \tan\theta_2 \Rightarrow \left(\frac{V}{T}\right)_1 < \left(\frac{V}{T}\right)_2$

From $PV = \mu RT$; $\frac{V}{T} \propto \frac{1}{P}$

Hence $\left(\frac{1}{P}\right)_1 < \left(\frac{1}{P}\right)_2 \Rightarrow P_1 > P_2$

Q.3 Answer:

a straight line passing through origin with a slope having different values for different gases

Solution:

$\frac{\mathrm{PV}}{\mathrm{T}} = \mathrm{nR} = \left(\frac{\mathrm{m}}{\mathrm{M}}\right)\mathrm{R}$ **or** $\frac{\mathrm{PV}}{\mathrm{T}} = \left(\frac{\mathrm{R}}{\mathrm{M}}\right)\mathrm{m}$

i.e., $\frac{\mathrm{PV}}{\mathrm{T}}$ **versus** m **graph is straight line passing through origin with slope** $\mathrm{R/M}$**, i.e. the slope depends on molecular mass of the gas** M **and is different for different gases.**

Q.4 Answer:

$\mathrm{Pm/kT}$

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Solution:

We know that $\mathrm{PV} = \mathrm{nRT} = (\mathrm{m}/\mathrm{M})\mathrm{RT}$
where $\mathrm{M} =$ **Molecular weight. Now** $\mathrm{P} \times \left(\frac{\mathrm{m}}{\mathrm{d}}\right) = \left(\frac{\mathrm{m}}{\mathrm{M}}\right)\mathrm{RT}$
where $\mathrm{d} =$ **density of the gas** $\frac{\mathrm{P}}{\mathrm{d}} = \frac{\mathrm{RT}}{\mathrm{M}} = \frac{\mathrm{kN_A T}}{\mathrm{M}} \quad \ldots$
where $\mathrm{R} = \mathrm{kN_A}, \mathrm{k}$ **is Boltzmann constant. But** $\frac{\mathrm{M}}{\mathrm{N_A}} = \mathrm{m} =$ **mass of each molecule so** $\mathrm{d} = \frac{\mathrm{P} \times \mathrm{m}}{\mathrm{kT}}$

Q.5 Answer:

$1\ \mathrm{g}$

Solution:

$\mathrm{PV} = \frac{\mathrm{m}}{\mathrm{M}}\mathrm{RT}$
Initially, $\mathrm{PV} = \frac{6}{\mathrm{M}}\mathrm{R} \times 500$
Finally, $\frac{P}{2}V = \frac{(6-x)}{M}R \times 300$ **(if** x **g gas leaks out** $)$ **Hence,** $2 = \frac{6}{6-\mathrm{x}} \times \frac{5}{3} \quad \therefore \mathrm{x} = 1\mathrm{gram}$

Q.6 Answer:

At low pressure and high temperature

Solution:

Real gas approach the ideal gas behaviour at low pressure and high temperature.

Q.7 Answer:

$\frac{4v}{\pi}$

Solution:

Figure shows the particles each moving with same spped v **but the different directions. Consider any two particles having angle** θ **between directions of their velocities.**
Then, $\overrightarrow{v_{rel}} = \overrightarrow{v_B} - \overrightarrow{v_A}$
i.e., $v_{\mathrm{rel}} = \sqrt{v^2 + v^2 - 2vv\cos\theta}$
$\Rightarrow v_{\mathrm{rel}} = \sqrt{2v^2(1-\cos\theta)} = 2v\sin(\theta/2)$
So averaging v_{rel} **over all pairs** $\bar{v}_{rel} = \frac{\int_0^{2\pi} v_{rel}\,d\theta}{\int_0^{2\pi} d\theta} = \frac{\int_0^{2\pi} 2v\sin(\theta/2)}{\int_0^{2\pi} d\theta}$
$= \frac{2v \times 2\left[-\cos(\theta/2)\right]_0^{2\pi}}{2\pi}$
$\Rightarrow \bar{v}_{rel} = (4v/\pi) > v[\mathrm{as}\ 4/\pi > 1]$

Q.8 Answer:

$\mathrm{V_1} = \mathrm{V_2},\ \mathrm{V_3} = \mathrm{V_4}$
and $\mathrm{V_2} > \mathrm{V_3}$

Solution:

From ideal gas equation $PV = \mu RT \Rightarrow$ **Slope of** $P.T$ **curve** $\frac{P}{T} = \frac{\mu R}{V}$
$\Rightarrow$ **Slope** $\propto \frac{1}{V}$
It means line of smaller slope represent greater volume of gas. For the given problem figure Point 1

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and 2 are on the same line so they will represent same volume i.e. $V_1 = V_2$ Similarly point 3 and 4 are on the same line so they will represent same volume i.e. $V_3 = V_4$ Also slope of line $1 - 2$ is less than $3 - 4$. Hence $\left(\hat{V}_1 = V_2\right) > (V_3 = V_4)$

Q.9 Answer:

$1 : 1$

Solution:

Vapour pressure does not depend on the amount of substance. It depends on the temperature alone.

Q.10 Answer:

$\mathrm{T_2 > T_1}$

Solution:

From ideal gas equation,

$\mathrm{PV = nRT}$

$\mathrm{P = \frac{wRT}{MV} = \frac{dRT}{M}}$

$\mathrm{PM = dRT}$

$\mathrm{\therefore P \propto d \cdot T}$

$\mathrm{\frac{d}{P}}$ **is inversely proportional to** T**.**

So $\mathrm{T_2 > T_1}$

Q.11 Answer:

isothermal process.

Solution:

According to boyle's law, at constant temperature. $\mathrm{P \propto \frac{1}{V}}$ **or** $\mathrm{P_1\,V_1 = P_2\,V_2}$

Q.12 Answer:

$2700\ \mathrm{m^3}$

Solution:

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$\therefore V_2 = \frac{P_1V_1}{T_1} \times \frac{T_2}{P_2} = \frac{4 \times 1500 \times 270}{300 \times 2} = 2700\ \mathrm{m^3}$

Q.13 Answer:

$T_1 < T_2$

Solution:

With increase in temperature, the product (PV) increases.

Q.14 Answer:

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$\frac{283}{383}x$

Solution:

Let the mass of the gas be m. At a fixed temperature and pressure, volume is fixed. Density of the gas, $\rho = \frac{\mathrm{m}}{\mathrm{V}}$ Now $\frac{\rho}{\mathrm{P}} = \frac{\mathrm{m}}{\mathrm{PV}} = \frac{\mathrm{m}}{\mathrm{nRT}}$

$\Rightarrow \frac{\mathrm{m}}{\mathrm{nRT}} = x$ (By question)

$\Rightarrow \mathrm{xT} =$ constant $\Rightarrow \mathrm{x}_1\ \mathrm{T}_1 = \mathrm{x}_2\ \mathrm{T}_2$

$$\Rightarrow \mathrm{x}_2 \Rightarrow \frac{\mathrm{x}_1\ \mathrm{T}_1}{\mathrm{T}_2} = \frac{283}{383}\mathrm{x} \left[\begin{array}{l} \therefore \\ \mathrm{T}_1 \\ \mathrm{T}_2 = 283\ \mathrm{K} \\ \mathrm{T} = 383\ \mathrm{K} \end{array}\right]$$

Q.15 Answer:

$2.7 \times 10^{-7}\ \mathrm{m}^3$

Solution:

We have $0.25 \times 6 \times 10^{23}$ molecules, each of volume $10^{-30}\ \mathrm{m}^3$.

Molecular volume $= 2.56 \times 10^{-7}\ \mathrm{m}^3$

Assuming ideal gas law is valid:

Final volume $= (\mathrm{V}_{\mathrm{in}}/100) = \left((3)^3 \times 10^{-6}\right)/100 \approx 2.7 \times 10^{-7}\ \mathrm{m}^3$

Q.16 Answer:

8 atmosphere

Solution:

$P_2 = P_1 \frac{V_1}{V_2}^{\gamma}$

(Adiabatic because sudden process)

$r = \frac{3}{2}, \quad V_2 = \frac{V_1}{4}, \quad P_1 = 1 \quad \text{atm}$

$P_2 = 1\left(4\right)^{3/2} = 8 \quad \text{atm}$

Q.17 Answer:

Remains the same

Solution:

The centre of mass of the gas molecules moves with uniform speed along with truck. As there is no change in relative motion, the translational kinetic energy and hence the temperature of the gas molecules will remain constant.

Q.18 Answer:

Remains constant.

Solution:

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Remains constant

Q.19 Answer:

1.67

Solution:

$U = \frac{f}{2}nRT$

In case of helium, being monoatomic gas, $f = 3$

$\therefore \quad U_{He} = \frac{3}{2}n_1RT$

For carbon monoxide, being diatomic gas, $f = 5$

$\therefore \quad U_{CO} = \frac{5}{2}n_2RT$

$\therefore \quad \frac{U_{He}}{U_{co}} = \frac{3}{5} \times \frac{n_1}{n_2}$

but $U_{He} = U_{CO}$

$\therefore \quad \frac{n_1}{n_2} = \frac{5}{3} = 1.67$

Q.20 Answer:

00.44

Solution:

Mean free path of a gas molecule,

$\lambda = \frac{1}{\sqrt{2}\pi d^2 n}$

As all the physical conditions are identical,

$\lambda \propto \frac{1}{d^2}$

$\therefore \quad \frac{\lambda_1}{\lambda_2} = \frac{d_2}{d_1}^2 = \frac{(1 \times 2)\backslash AA}{(1{:}5 \times 2)\backslash AA}^2$

$= \frac{1}{3/2}^2$

$= \frac{4}{9}$

$\therefore \quad \frac{\lambda_1}{\lambda_2} = 0.44$

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